

PAULEE GROUP · CENTER FOR QUANTUM TECHNOLOGY, KIST

Convolution Operation and Quantum Signal Processing

Donghun Jung

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Outline

1. Introduction to Quantum Information
2. Convolution Operation
3. Quantum Signal Processing

INTRO

METHOD

STRATEGY

RESULTS

ONGOING

01

Introduction to Quantum Information

Everything we do is three steps

Quantum information is an enormous field, but what we actually run in the lab generally has the same three steps.

STEP ONE

Prepare a state

The system is initialised into one known starting state.



STEP TWO

Apply a unitary

A quantum operation, which is to say an algorithm, evolves the state the way we want.



STEP THREE

Measure

One number comes back, and every conclusion has to be extracted from it.

$$|\psi_0\rangle \xrightarrow{U} U|\psi_0\rangle,$$

$$\text{what we read out} = \langle\psi_0|U^\dagger O U|\psi_0\rangle.$$

Each box has a long story behind it, and each is a field of its own. On the theory side the main interest sits in the unitary U : it is the part we choose, and we design it on purpose so that the number produced by the third box answers the question we asked.

State preparation

A state of a d -level system is a ket, a column of complex amplitudes in a basis $\{|j\rangle\}$:

$$|\psi\rangle = \sum_{j=0}^{d-1} c_j |j\rangle = \begin{pmatrix} c_0 \\ c_1 \\ \vdots \\ c_{d-1} \end{pmatrix} \quad c_j \in \mathbb{C}.$$

The bra is its Hermitian conjugate, the row vector of conjugated amplitudes $\langle\psi| = (c_0^*, \dots, c_{d-1}^*)$, and the norm is the bra taken against the ket,

$$\langle\psi|\psi\rangle = \sum_j |c_j|^2 = 1,$$

which is the normalization condition.

Every quantity we can actually observe is a bra times a ket: an expectation value $\langle\psi|O|\psi\rangle$, or an overlap $|\langle\phi|\psi\rangle|^2$.

Multiply the ket by a phase, $|\psi\rangle \rightarrow e^{i\alpha}|\psi\rangle$, and the bra picks up the conjugate, $\langle\psi| \rightarrow e^{-i\alpha}\langle\psi|$, so every such product is unchanged:

$$\langle\psi|e^{-i\alpha} O e^{i\alpha}|\psi\rangle = \langle\psi|O|\psi\rangle, \quad |\langle\phi|e^{i\alpha}|\psi\rangle|^2 = |\langle\phi|\psi\rangle|^2.$$

Therefore we lose no generality by taking the first amplitude c_0 to be real and non-negative.

So a d -level state needs $2d - 2$ real numbers: $2d$ from the complex amplitudes, minus one for the norm and one for the phase. A qubit then carries 4 real numbers, minus those two, leaving **two**, which is why the next slides draw it as a point on a sphere.

Expectation value

An observable is a Hermitian matrix $O = O^\dagger$, and the number the experiment reports is $\langle O \rangle_\psi = \langle \psi | O | \psi \rangle = \sum_{j,k} c_j^* O_{jk} c_k$, a row vector times a matrix times a column vector, which Hermiticity makes real.

For a single qubit the observables of interest are the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and we write the row, the matrix and the column, then multiply.

With $|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ this gives

$$\langle + | X | + \rangle = \frac{1}{2} (1, 1) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1.$$

One shot gives an eigenvalue, many shots give the average

The trace is cyclic, so the same number can be written with the state moved to the outside:

$$\langle \psi | O | \psi \rangle = \text{Tr} (| \psi \rangle \langle \psi | O) = \text{Tr}(\rho O), \quad \rho = | \psi \rangle \langle \psi |.$$

The object ρ is the **density matrix**, which is what one half of an entangled pair becomes later in this chapter.

A single shot does not return the average.

Diagonalise the observable, $O = \sum_k \lambda_k |k\rangle \langle k|$. One shot returns **one eigenvalue** λ_k with the Born probability $p_k = | \langle k | \psi \rangle |^2$, and it leaves the system in the corresponding eigenvector $|k\rangle$, which is the collapse.

Averaging those outcomes gives

$\sum_k \lambda_k p_k = \langle \psi | O | \psi \rangle$: the expectation value **is** the shot-averaged outcome, reached with an error of order $1/\sqrt{N}$ after N shots. For Z the eigenvalues are ± 1 and $\langle Z \rangle = p_0 - p_1$.

Bloch sphere

A qubit state is a vector in a two-dimensional complex space, $|\psi\rangle = c_0|0\rangle + c_1|1\rangle$ with $|c_0|^2 + |c_1|^2 = 1$.

Take the expectation value of each Pauli matrix, $\langle\sigma_i\rangle = \langle\psi|\sigma_i|\psi\rangle$, so the three form the **Bloch vector**:

$$\vec{r} = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle).$$

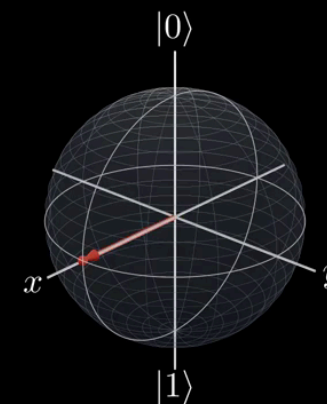
For example, for $|+\rangle$,

$$\langle X \rangle = \frac{1}{2} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1,$$

with $\langle Y \rangle = \langle Z \rangle = 0$, so $|+\rangle$ sits at $\vec{r} = (1, 0, 0)$.

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\vec{r} = (\langle X \rangle, \langle Y \rangle, \langle Z \rangle) = (1, 0, 0)$$



The three products for $|+\rangle$ written out in numbers, the matrix times the column first, then the row times that column; the arrow then appears at $\vec{r} = (1, 0, 0)$.

The two angles are the sphere's own coordinates

A ket has four real numbers. Normalisation removes one, and the global phase, which no expectation value can see, removes another. So only two real parameters are left, and by convention we represent a ket state as

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle.$$

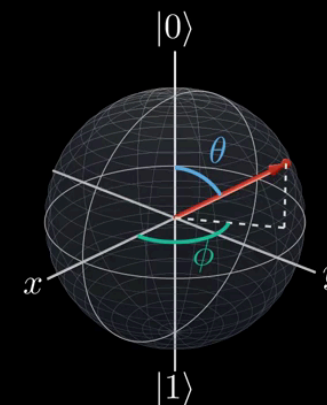
Feeding this into $\langle X \rangle = 2 \operatorname{Re}(c_0^* c_1)$,
 $\langle Y \rangle = 2 \operatorname{Im}(c_0^* c_1)$ and $\langle Z \rangle = |c_0|^2 - |c_1|^2$ gives

$$\vec{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta).$$

Those are the spherical coordinates of a unit vector, which is why this parametrisation is used. θ is the polar angle from $|0\rangle$, ϕ the azimuth around z , and the half angle inside the ket is what makes the full angle θ appear on the sphere. Since $|\vec{r}|^2 = \sin^2 \theta + \cos^2 \theta = 1$, **every ket sits on the surface**. Note that the radius is the purity $|\vec{r}|^2 = 2 \operatorname{Tr} \rho^2 - 1$.

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle = \begin{pmatrix} \cos \frac{\theta}{2} \\ e^{i\phi} \sin \frac{\theta}{2} \end{pmatrix}$$

$$\vec{r} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$$



$$\theta = 58^\circ$$

$$\phi = 120^\circ$$

The same products for the general ket, then the arrow on the sphere with the polar angle θ (blue arc from the z axis) and the azimuth ϕ (green arc from the x axis) following it as the two angles are swept.

Our case: prepare $|0\rangle$, read out along z



A laser pulse pumps the spin into $|0\rangle$, the unitary we designed acts, and a second laser pulse returns a photon count.

Strictly, the last step is a projective measurement on $|0\rangle$ rather than a measurement of Z , because a photon count is intrinsically positive. The observable is the projector $\Pi_0 = |0\rangle\langle 0|$, not an operator with eigenvalues ± 1 . The two are one affine step apart, since $\Pi_0 = \frac{1}{2}(I + Z)$:

$$P(0) = \langle \psi | \Pi_0 | \psi \rangle = \frac{1 + \langle Z \rangle}{2}, \quad \langle Z \rangle = 2P(0) - 1.$$

The mean count is linear in $P(0)$ between a bright and a dark reference level, so dividing by that contrast turns photons into $P(0)$. **After contrast normalisation the readout is effectively a Z measurement**, and $\langle Z \rangle$ is what every curve in this talk plots.

Schrodinger equation

The state obeys the Schrodinger equation (with $\hbar = 1$),

$$\frac{d}{dt}|\psi(t)\rangle = -i H |\psi(t)\rangle,$$

where H is the Hamiltonian, the matrix whose expectation value is the energy, and it is Hermitian, $H^\dagger = H$, because energies are real. If H does not depend on time this is a linear equation with constant coefficients, so an exponential solves it,

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle, \quad U(t) = e^{-iHt} = \sum_{n=0}^{\infty} \frac{(-iHt)^n}{n!}.$$

It is easy to see that $U^\dagger U = e^{+iH^\dagger t} e^{-iHt} = e^{+iHt} e^{-iHt} = I$, and the same in the other order. So the Schrodinger equation generates a unitary operation, and the norm of the state is preserved.

Hamiltonian and unitary operation

To implement the gate operation we need, we have to engineer the Hamiltonian. The Hamiltonian is always drift plus control, $H(t) = H_0 + H_c(t)$: what the system does on its own, plus the pulse we add.

For example, the NV centre has its own Hamiltonian, and we can apply a microwave pulse to it, which is effectively an oscillating magnetic field:

$$\begin{aligned} H_0 &= D S_z^2 + \gamma B_{\parallel} S_z, \\ H_c(t) &= \Omega(t) \cos(\omega t + \varphi) S_x. \end{aligned}$$

Here D is the zero-field splitting, $\gamma B_{\parallel}$ the Zeeman shift, $\Omega(t)$ the drive amplitude, ω its carrier and φ its phase, which in the rotating frame selects the rotation axis; the sequences later use $\varphi = \pi/2$, a y drive.

Now $H(t_1)$ and $H(t_2)$ no longer commute, and the time-dependent Schrodinger equation has no closed form in general; solving it efficiently is a challenge of its own and out of scope for today. Briefly, one can step it forward like the Euler method,

$$|\psi(t + dt)\rangle = |\psi(t)\rangle - i H(t) |\psi(t)\rangle dt,$$

and the product of many such small steps is the unitary U .

Hamiltonian on Bloch Sphere

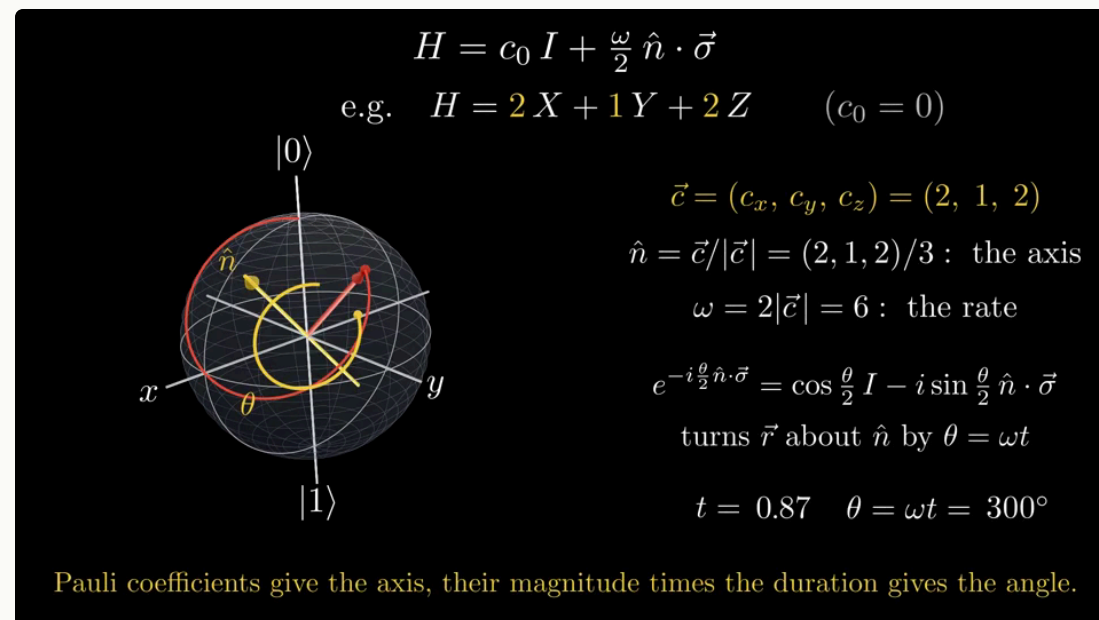
A 2×2 Hermitian matrix has only four real parameters, so it can always be written as a linear combination of Pauli matrices,

$$H = c_0 I + \frac{\omega}{2} \hat{n} \cdot \vec{\sigma}, \quad \hat{n} = \frac{\vec{c}}{|\vec{c}|}, \quad \omega = 2|\vec{c}|.$$

Here $\vec{c} = (c_x, c_y, c_z)$ collects the Pauli coefficients and $c_0 I$ is only a global phase. It is useful to use

$$e^{-i\frac{\theta}{2}\hat{n}\cdot\vec{\sigma}} = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \hat{n} \cdot \vec{\sigma}, \quad \theta = \omega t.$$

This rotates the Bloch vector about the axis $\hat{n}$ by the angle θ . The Pauli coefficients give the axis, their magnitude times the duration gives the angle. A resonant pulse, $H = \frac{\Omega}{2} Y$, turns the state about y by Ωt ; free precession, $H = \frac{\delta}{2} Z$, turns it about z by δt . A pulse sequence is now a list of axes and angles.



The example $H = 2X + Y + 2Z$ has $\vec{c} = (2, 1, 2)$, so the axis is $\hat{n} = (2, 1, 2)/3$ and the angle grows as $\theta = 6t$; the state $|0\rangle$ precesses on a cone about $\hat{n}$. Then the two cases of the next slides: $\frac{\Omega}{2} Y$ turns $|0\rangle$ toward $|+\rangle$ by Ωt , and $\frac{\delta}{2} Z$ turns $|+\rangle$ about z by δt .

So, prepare $|0\rangle$, apply a pulse U , and read $P(0)$.

Take one spin-1 transition as the qubit. In the rotating frame, after the rotating-wave approximation and with the drive phase along y , the Hamiltonian is constant:

$$H = \frac{\delta}{2} Z + \frac{\Omega}{2} Y.$$

The detuning δ is where the field enters and $\Omega = \gamma B_1$ is the drive strength.

The exponent is written in the Pauli basis, so the time evolution is a rotation and both of its parts are read off the coefficients:

$$U = e^{-iHt} = \exp\left(-\frac{i}{2}\Omega_R t \hat{n} \cdot \vec{\sigma}\right), \quad \hat{n} = \frac{(0, \Omega, \delta)}{\Omega_R}.$$

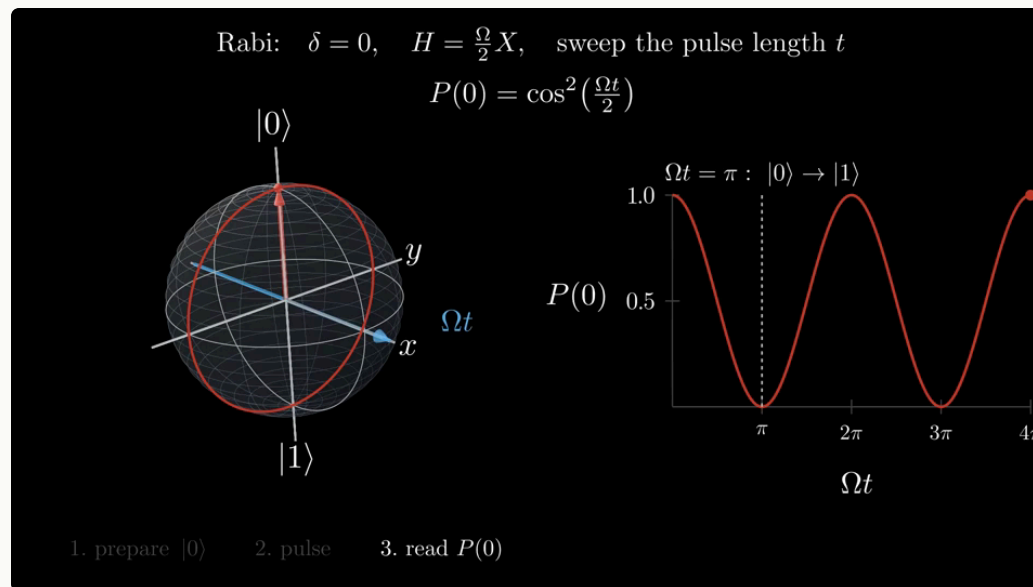
The **axis** is the coefficient vector, the **angle** is its length $\Omega_R = \sqrt{\Omega^2 + \delta^2}$ times the time. Drive off means precession about z .

Rabi: calibrating the rotation angle

Setting the pulse on resonance, $\delta = 0$, so $H = \frac{\Omega}{2}Y$ and the pulse is a rotation about y by the angle Ωt :

$$P(0) = |\langle 0 | e^{-i\Omega Y t/2} | 0 \rangle|^2 = \cos^2\left(\frac{\Omega t}{2}\right).$$

Sweep the pulse length and the fluorescence oscillates at Ω .



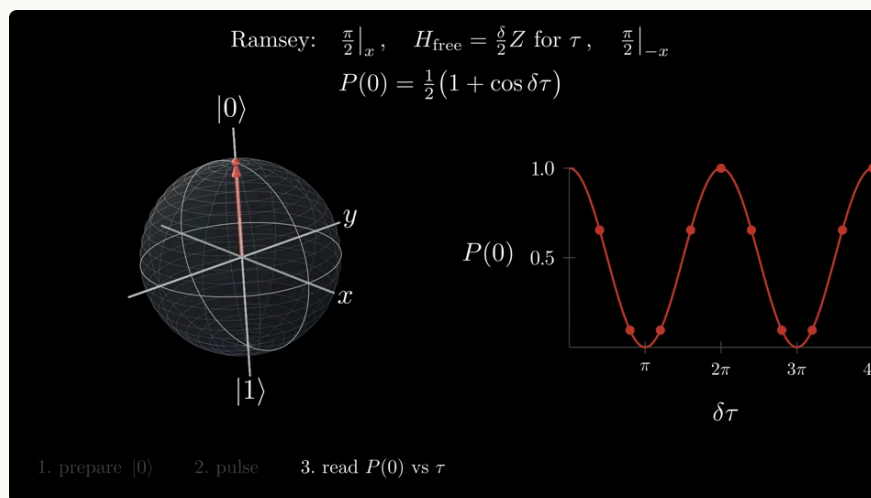
Each pulse length is a fresh run: rotate about y by Ωt , project onto z , and plot $P(0)$.

Ramsey

A $\pi/2$ pulse about the y axis gives the $|+\rangle$ state. Then, with the drive off, $\Omega = 0$, only $H = \frac{\delta}{2}Z$ acts, so the state precesses about z by $\delta\tau$. Then apply a second $\pi/2$ pulse, about $-y$; the resulting population depends on that phase:

$$P(0) = \frac{1}{2} \left[1 + \cos(\delta\tau) \right].$$

The frequency is the detuning itself. A field shift moves δ , and the fringe counts it.

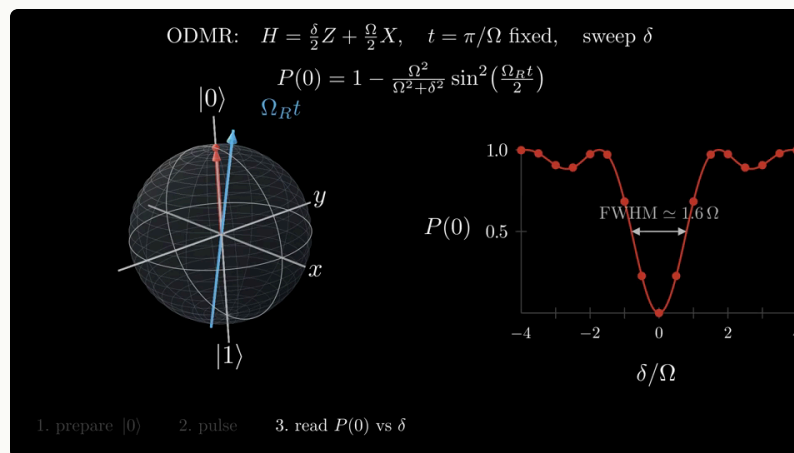


For each τ : $\pi/2$ about y , precession about z by $\delta\tau$, $\pi/2$ about $-y$, then the projection onto z gives one point.

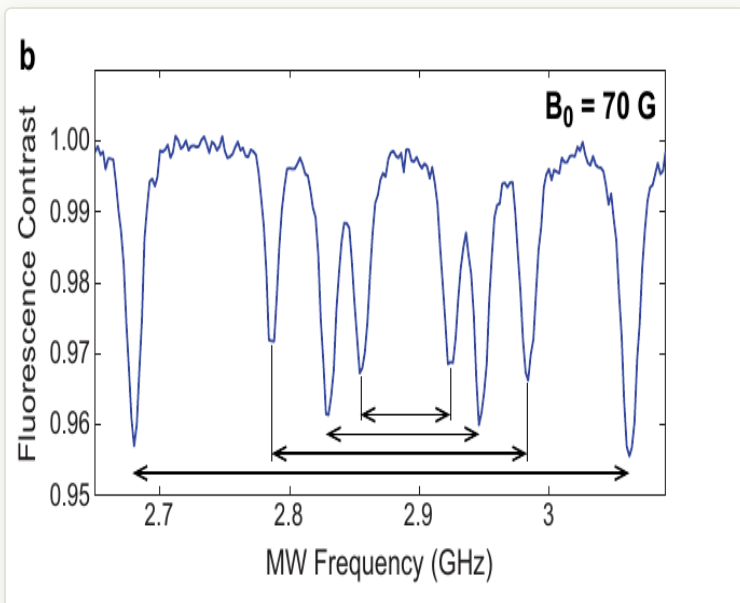
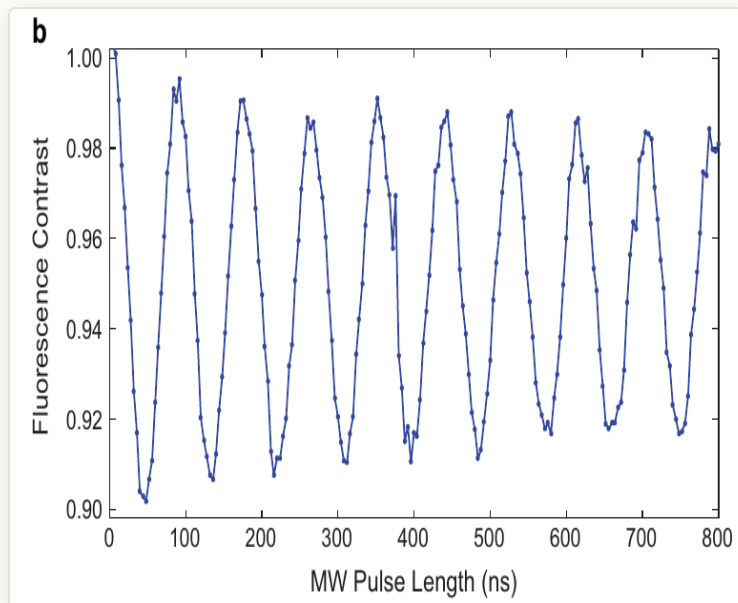
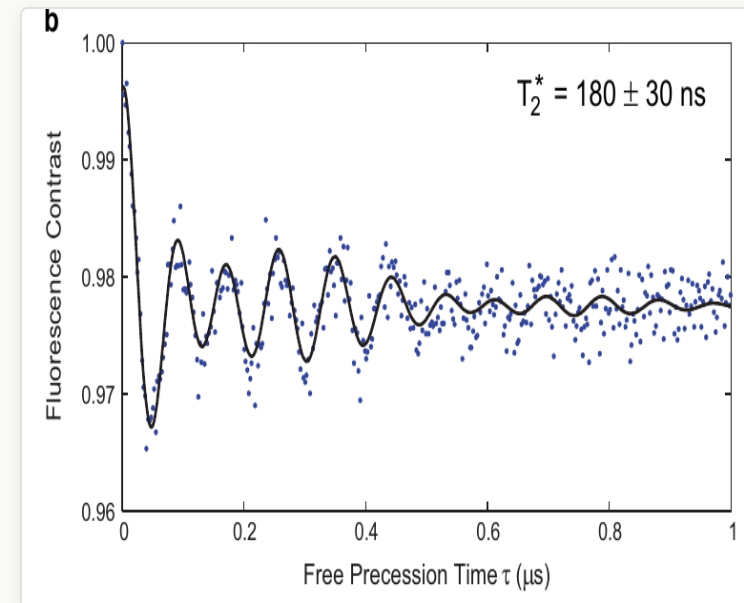
Fix the pulse at the calibrated length $t = \pi/\Omega$ and sweep the carrier instead. So on resonance, $\delta = 0$, we have $\Omega t = \pi$, a π pulse. Off resonance, the rotation axis $\hat{n} = (0, \Omega, \delta)/\Omega_R$ tilts toward z and the rotation is faster:

$$P(0) = 1 - \frac{\Omega^2}{\Omega^2 + \delta^2} \sin^2\left(\frac{\Omega_R t}{2}\right).$$

The resulting signal is approximately a sinc function.



The axis tilts as δ is stepped across $[-4\Omega, 4\Omega]$, and one π pulse per step draws out the dip and its side lobes.

*ODMR**Rabi**Ramsey*

Measured on an NV ensemble (Pham thesis, Ch. 1, Figs. 1.4, 1.5, 1.7). Two things remain to discuss. One is how the response is lost as time goes on, and the other is the source of the response.

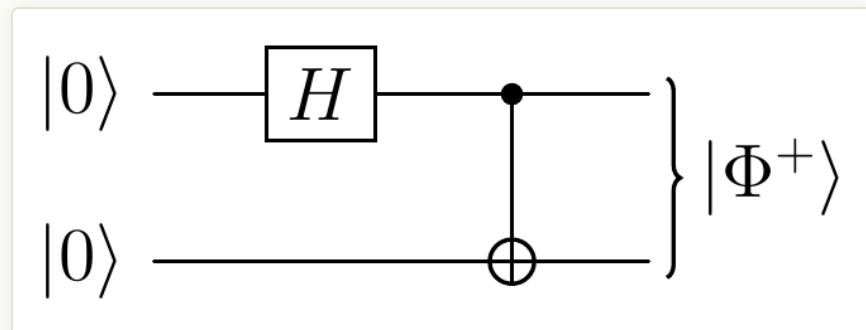
Entanglement

We start from a product state, $|\psi_0\rangle = |0\rangle|0\rangle$. A Hadamard acts on the first qubit and leaves the second one alone,

$$(H \otimes I) |00\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)|0\rangle.$$

The CNOT is **conditional**: it flips the second qubit only when the first one is $|1\rangle$. It gives the maximally entangled Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle).$$



Any product state can be written as $(a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle)$, which would need $ad = bc = 0$ together with $ac = bd \neq 0$. No product of two kets reproduces the Bell state.

Suppose our interest is the first qubit. Every measurement on it is $\text{Tr}(\rho_1 O)$, with the other qubits traced out,

$$\rho_1 = \text{Tr}_2 |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I.$$

That matrix has rank two, while a ket gives $\rho = |\psi\rangle\langle\psi|$ of rank one; no ket describes the first qubit alone, so we call it **mixed**.

Every Pauli matrix is traceless, so all three expectation values vanish, $\langle\sigma_i\rangle = \text{Tr}\left(\frac{1}{2}I\sigma_i\right) = \frac{1}{2}\text{Tr}\sigma_i = 0$, and the arrow sits at the centre, $|\vec{r}_1| = 0$. The length is the purity, $\text{Tr}\rho^2 = \frac{1}{2}(1 + |\vec{r}|^2)$: 1 for a ket on the surface, 1/2 at the centre.

Two qubits: an interaction builds the Bell state

$$|\psi_0\rangle = |0\rangle|0\rangle$$

$$(H \otimes I)|\psi_0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|0\rangle$$

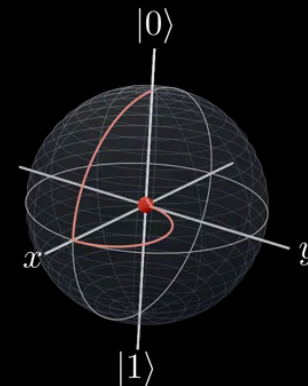
$$\text{CNOT}: |\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$\rho_1 = \text{Tr}_2 |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I$$

$$\langle\sigma_i\rangle = \text{Tr}\left(\frac{1}{2}I\sigma_i\right) = \frac{1}{2}\text{Tr}\sigma_i = 0 \Rightarrow |\vec{r}_1| = 0$$

$$\langle X \rangle = 0.00 \quad \langle Y \rangle = 0.00 \quad \langle Z \rangle = 0.00$$

$$|\vec{r}_1| = \sqrt{\langle X \rangle^2 + \langle Y \rangle^2 + \langle Z \rangle^2} = 0.00$$



The reduced Bloch vector of qubit 1: from $|0\rangle$ to $|+\rangle$ under the Hadamard, then to the origin during the CNOT, with $\langle X \rangle$, $\langle Y \rangle$, $\langle Z \rangle$ read off the partial trace.

Entanglement with the outside looks like mixing inside our qubit: the more entangled, the shorter the Bloch vector.

Interaction is the source of entanglement.

The CNOT came from an interaction, and in the NV centre the hyperfine coupling to a nearby nucleus has the same conditional form, with $\hat{H}_0 = \omega_L \hat{I}_z$ and $\hat{H}_1 = \hat{H}_0 + A\hat{I}_z + B\hat{I}_x$:

$$\begin{aligned}\hat{H} &= |0\rangle\langle 0| \otimes \hat{H}_0 + |1\rangle\langle 1| \otimes \hat{H}_1, \\ |+\rangle|n\rangle &\rightarrow \frac{1}{\sqrt{2}} (|0\rangle U_0 |n\rangle + |1\rangle U_1 |n\rangle), \\ |\vec{r}_e| &= |\langle n| U_1^\dagger U_0 |n\rangle|.\end{aligned}$$

The nucleus precesses about a different axis in each branch, since $\hat{H}_0 \neq \hat{H}_1$, so the electron's Bloch radius is the overlap of the two branches. There are many independent nuclei, so multiply their overlaps, giving $M(t) \simeq e^{-(t/T_2^*)^2}$.

One nucleus with $\omega_L = 1$, $A = 0.5$, $B = 1$ in scene units: branch 0 precesses about $\hat{n}_0 = \hat{z}$, branch 1 about $\hat{n}_1$ tilted by 34° toward x , and the radius of their average collapses and revives. Five nuclei at random couplings then give the Gaussian decay (conditional form from Taminiau et al., 2012).

The decay has a name: T_2^*

The shrinking radius of the last slide is exactly what a Ramsey experiment measures. The free-precession signal fades within a time T_2^* , set by the static spread of local fields: each nucleus shifts the electron's precession frequency by its own hyperfine coupling, and the many shifted precessions dephase when we add them up,

$$M(t) \simeq e^{-(t/T_2^*)^2}, \quad T_2^* \approx 180 \text{ ns for a single NV}, \quad T_2^* \approx 500 \text{ ns for an ensemble}.$$

Both numbers are for NV centres at room temperature (Pham thesis, Ch. 1). The static field comes mostly from the ^{13}C bath at its natural abundance of 1.1%, which caps T_2^* near $1 \mu\text{s}$; isotopically purified 99.7% ^{12}C material exceeds $10 \mu\text{s}$.

Dynamical decoupling: make the conditional operation unconditional

Write the joint Hamiltonian in the electron basis and exponentiate it block by block,

$$H = |0\rangle\langle 0| \otimes H_0 + |1\rangle\langle 1| \otimes H_1 \quad \Longrightarrow \quad U(t) = e^{-iHt} = |0\rangle\langle 0| \otimes U_0(t) + |1\rangle\langle 1| \otimes U_1(t).$$

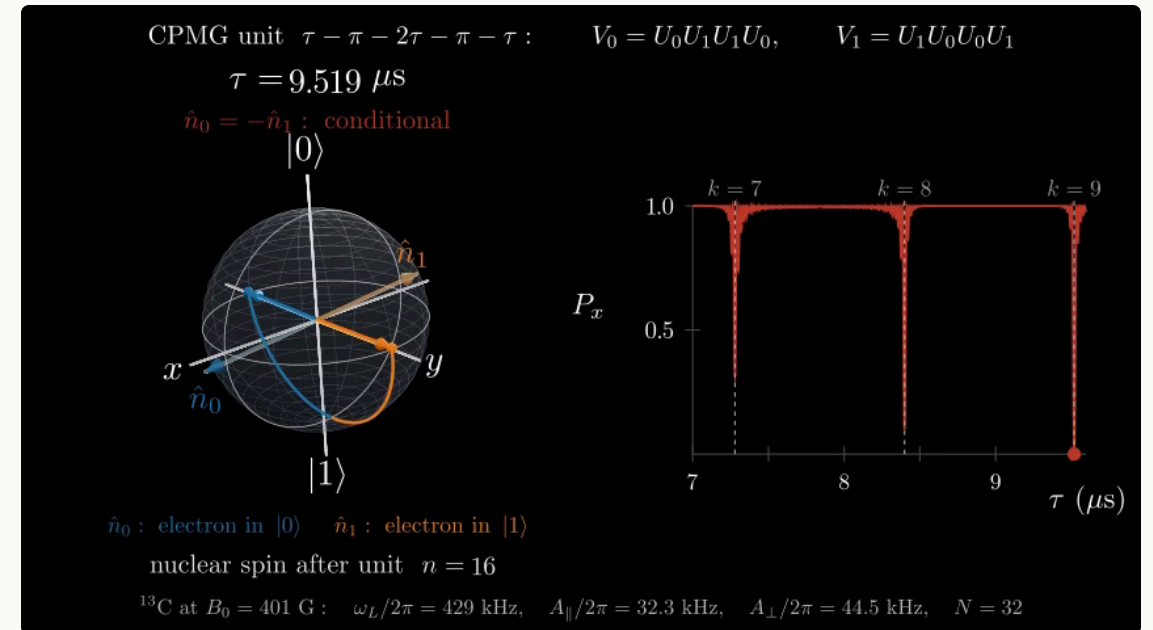
For one ^{13}C near an NV centre,

$$H_0 = \omega_L I_z \text{ and } H_1 = (\omega_L + A_{\parallel}) I_z + A_{\perp} I_x.$$

A π pulse swaps the two blocks, so the CPMG unit $\tau - \pi - 2\tau - \pi - \tau$ interleaves them, and where the electron started decides which comes first.

$$V_0 = U_0 U_1 U_1 U_0, \quad V_1 = U_1 U_0 U_0 U_1, \quad U_b \equiv U_b(\tau).$$

Both are unitary, so the only question is whether the two rotations agree.



One unit at the first resonance, $\tau = \tau_1 = 0.561 \mu\text{s}$, for the nuclear spin with the electron in $|0\rangle$ (left) and in $|1\rangle$ (right). Every factor is a rotation about the active block's axis, then the four factors collapse into one rotation per branch by the same ϕ about opposite axes $\hat{n}_0$ and $\hat{n}_1$.

The read-out is one number: how well the two branches agree

Prepare the electron in $|+\rangle$, leave the nucleus unpolarised, $\rho_n = I/2$, and run N units:

$$\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \rho_n \longrightarrow \frac{1}{\sqrt{2}}(|0\rangle \otimes V_0^N + |1\rangle \otimes V_1^N) \text{ acting on } \rho_n.$$

Tracing the nucleus out leaves the electron coherence

$$\langle 0 | \rho_e | 1 \rangle = \frac{1}{2} \text{Tr}_n (V_0^N \rho_n V_1^{\dagger N}) = \frac{1}{2} M,$$

so a read-out along x returns

$$P_x = \frac{1}{2} + \text{Re} \langle 0 | \rho_e | 1 \rangle = \frac{1 + M}{2}, \quad M = \frac{1}{2} \text{Tr} (V_0^N V_1^{\dagger N}).$$

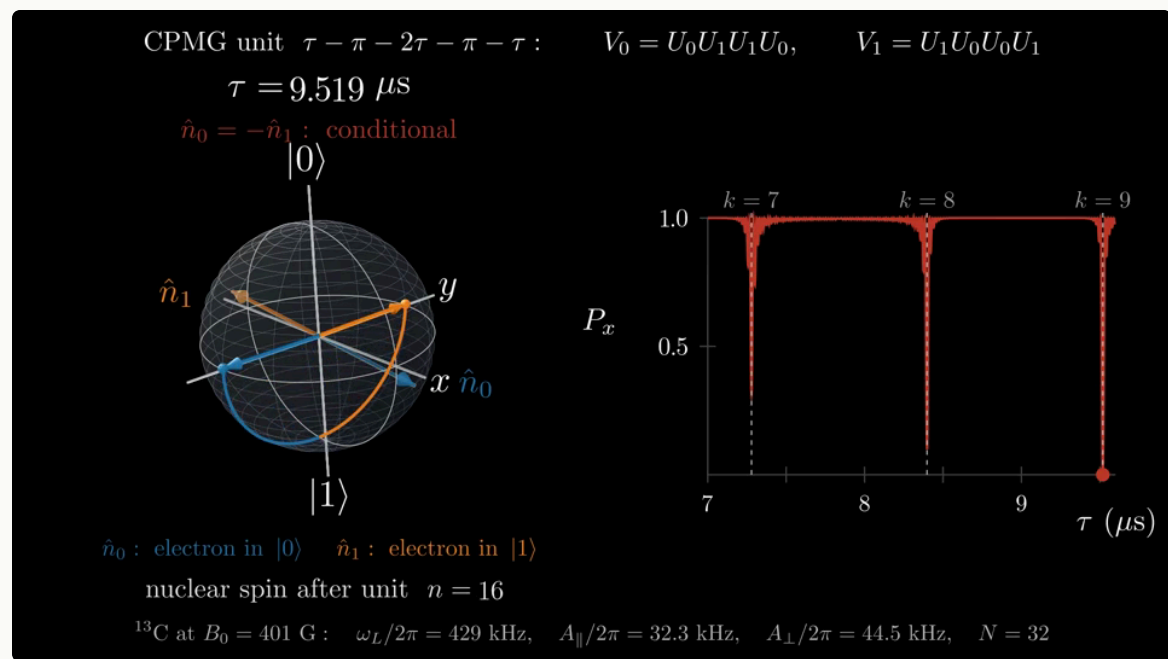
That $\frac{1}{2}$ is the trace normalisation of the unpolarised nucleus, so M is an average of $V_1^{\dagger N} V_0^N$ and $|M| \leq 1$. If the operation is **unconditional**, $V_0 = V_1$, then $M = 1$ exactly.

The electron Bloch vector has length $|M|$ and purity $\text{Tr} \rho_e^2 = \frac{1}{2}(1 + |M|^2)$, so staying unconditional keeps the state pure and the signal full. That is why decoupling buys a coherence time $T_2^{(N)}$ far longer than the single-pulse echo T_2 (Pham thesis, Ch. 1).

Same angle, different axis: the dips that name one nucleus

With $A = U_0(\tau)$, $B = U_1(\tau)$, cyclicity gives $\text{Tr } V_0 = \text{Tr}(ABBA) = \text{Tr}(A^2B^2) = \text{Tr } V_1$, so both branches turn by the **same angle** ϕ and only the axes $\hat{n}_0, \hat{n}_1$ can differ; each nucleus has its own $A_{\parallel}$, hence its own comb τ_k :

$$M = 1 - (1 - \hat{n}_0 \cdot \hat{n}_1) \sin^2 \frac{N\phi}{2}, \quad \tau_k = \frac{(2k-1)\pi}{2\omega_L + A_{\parallel}}.$$



Spin 3 of the paper's Table I ($B_0 = 401 \text{ G}$, $N = 32$ units): the axes coincide except at τ_7, τ_8, τ_9 , where they anti-align and M plunges.

What to carry out of chapter 1

- **An experiment is three steps.** We prepare a known state, apply a unitary, and measure.
- **The unitary is the part we design on purpose.**
- **The response is not the sequence alone.** It also carries the configuration of the system.

The measured curve is the designed response combined with the structure of the system, and that combination is a convolution.

02

Convolution Operation

The response is already convolved

We prepare a state, apply a unitary we designed, and measure. The outcome can be read as how the source, in our case the quantum system, responds to our input, the signal. I prefer to call that output the **response**, spectrum included, and it is convolved: it carries the configuration of the system as well as the sequence we ran.

Sweep a knob ω and the curve on the screen is never the bare spectrum $S(\omega)$ of the sample. It is

$$S_{\text{meas}}(\omega) = \int d\omega' S(\omega') K(\omega - \omega') \equiv [S * K](\omega),$$

which is the definition of a convolution. The definition looks complicated, so before using it let me show why convolution is natural, starting from two dice.

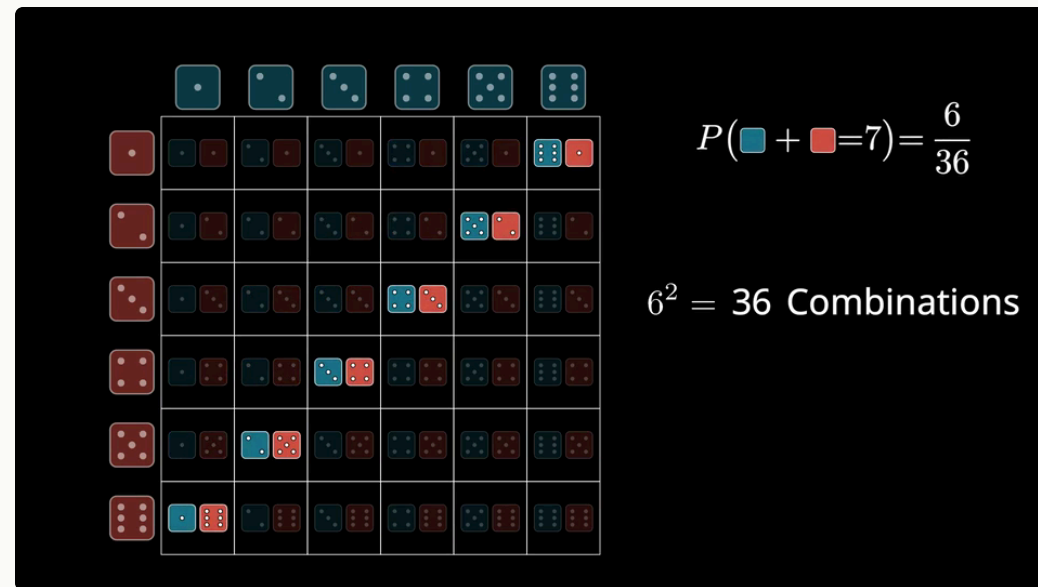
Two dice: the table of outcomes

Roll two independent dice X and Y and ask for the probability of their sum, $Z = X + Y$. The natural picture is a table: 36 cells, one per pair (X, Y) , each with weight $\frac{1}{36}$.

The event $Z = n$ happens whenever $X = k$ and $Y = n - k$, for any k , so the cells with a fixed sum lie on one anti-diagonal of the table:

$$P(Z = n) = \sum_k P(X = k) P(Y = n - k).$$

Counting the anti-diagonal for $n = 7$ gives six cells, so $P(Z = 7) = 6/36$, the largest of all.



The 6×6 table of pairs, and the anti-diagonals $X + Y = n$ marched one by one. Each diagonal is one entry of the distribution of the sum.

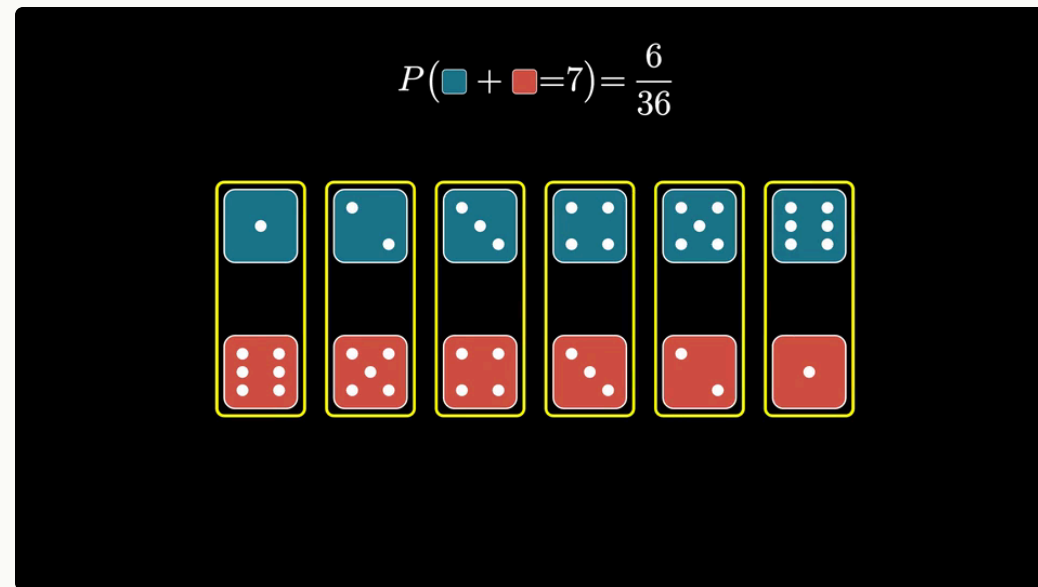
The same sum, seen as flip and slide

Write the two distributions as lists. Reverse one of them, slide it along the other, and at each displacement multiply the entries that face each other and add the products. That single number is $P(Z = n)$ for that displacement.

$$(f * g)[n] = \sum_k f[k] g[n - k].$$

The minus sign is the **flip** and n is the **slide**. The only rule is that the two indices add up to the output index, since $k + (n - k) = n$; everything else is bookkeeping.

Two lists of length 6 overlap at 11 displacements, the sums 2 to 12.



The second row is reversed and slid across the first. The pairs that line up are exactly the anti-diagonal of the table, now read as a moving window.

Uneven dice: the same picture with weights

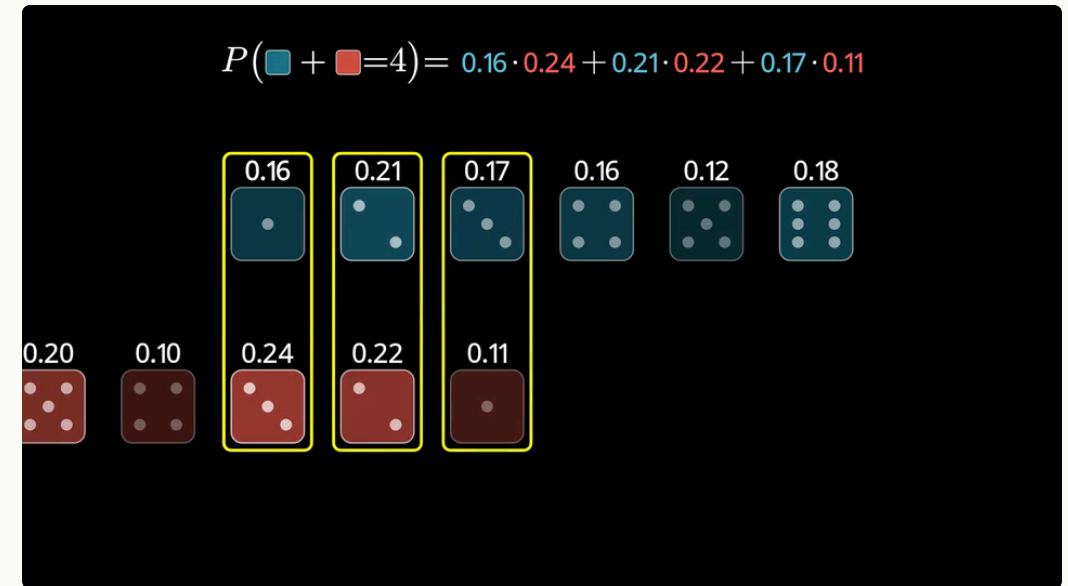
Now load the dice, so the faces are no longer equally likely. Nothing changes in the bookkeeping: the same anti-diagonal is summed, only each cell now carries the product of two unequal weights,

$$P(Z = n) = \sum_k P(X = k) P(Y = n - k) = [P_X * P_Y](n).$$

For example, the clip reads off

$$P(Z = 4) = 0.16 \cdot 0.24 + 0.21 \cdot 0.22 + 0.17 \cdot 0.11.$$

Nobody chose to convolve anything. Adding two independent random variables **is** a convolution of their distributions.



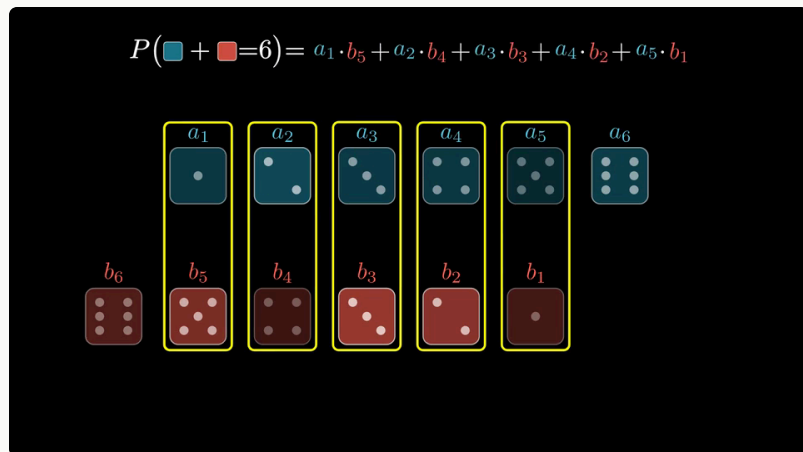
Loaded dice: the same table and the same anti-diagonals, now weighted cell by cell.

Convolution of (a_i) and (b_i)

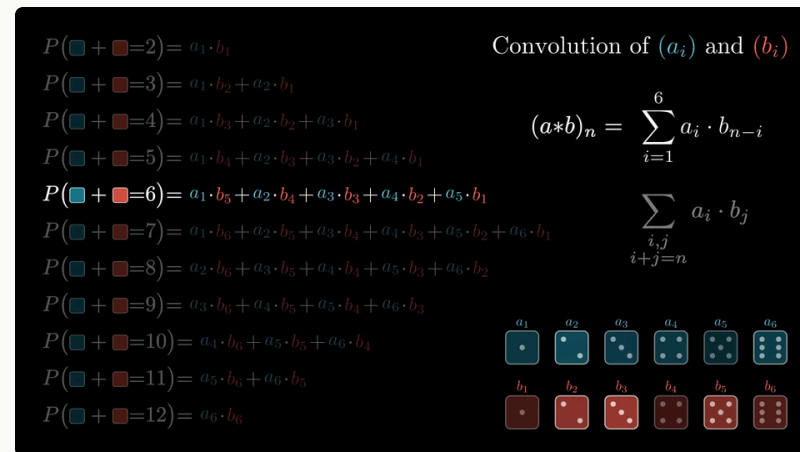
Replace the weights by symbols, $P_X(i) = a_i$ and $P_Y(j) = b_j$. The sliding window collects the pairs whose indices add up to n , and that is the whole definition:

$$(a * b)_n = \sum_{i+j=n} a_i b_j = \sum_i a_i b_{n-i}.$$

Up to now we have only added two independent random variables. The convolution was not a choice; it is what addition looks like in the distributions.



Symbolic weights, one window position per output entry.



The same products as a table, then the formula.

Forget the dice: any two lists convolve the same way

Take two plain lists, $(1, 2, 3)$ and $(4, 5, 6)$, with no probabilities attached. Reverse one, slide it, multiply what faces each other, add:

$$(1, 2, 3) * (4, 5, 6) = (4, 13, 28, 27, 18).$$

At displacement $n = 3$ the window holds $2 \cdot 6 + 3 \cdot 5 = 27$. Two lists of length 3 overlap at 5 displacements, so the output has length 5.

This is the discrete definition, and the continuous one is the same sentence with an integral in place of the sum:

$$(f * g)(x) = \int f(y) g(x - y) dy.$$

$$(1, 2, 3) * (4, 5, 6) = (4, 13, 28, \quad)$$

$$1 \cdot 6 + 2 \cdot 5 + 3 \cdot 4$$

a_0	a_1	a_2
1	2	3
6	5	4
b_2	b_1	b_0

One list reversed and slid across the other; each frame is one output entry.

A product of polynomials is a convolution of coefficients

Use the same numbers as coefficients and multiply two polynomials, which is an ordinary **pointwise** product of two functions of x :

$$\begin{aligned} & (1 + 2x + 3x^2)(4 + 5x + 6x^2) \\ &= 4 + 13x + 28x^2 + 27x^3 + 18x^4. \end{aligned}$$

The coefficient of x^n collects every pair of powers that adds up to n , and that is the flip-and-slide sum again:

$$c_n = \sum_{i+j=n} a_i b_j = (a * b)[n].$$

Multiplying by x^j **shifts** the coefficient list by j , so a product of polynomials is a weighted sum of shifted copies, which is what a convolution is.

$$(1, 2, 3) * (4, 5, 6) = (4, 13, 28, 27, 18)$$

$$(1 + 2x + 3x^2)(4 + 5x + 6x^2) = 4 + 13x + 28x^2 + 27x^3 + 18x^4$$

	1	$2x$	$3x^2$
4	4	$8x$	$12x^2$
$5x$	$5x$	$10x^2$	$15x^3$
$6x^2$	$6x^2$	$12x^3$	$18x^4$

Each power of x collects one diagonal of the table of pairwise products, and the diagonals are exactly the sums $i + j = n$.

One identity: convolve here, multiply there

$$\widehat{f * g} = \hat{f} \cdot \hat{g} \quad \Longleftrightarrow \quad \widehat{f \cdot g} = \hat{f} * \hat{g}.$$

A convolution treats all positions alike, so it commutes with shifts, and the complex exponentials $e^{i\omega x}$ are its eigenfunctions. In that basis the operation is **diagonal**, and diagonal means multiply entry by entry.

Read the other way, a product of two records in time is a convolution of their spectra, so a finite measurement window smears every line by the transform of that window.

A trigonometric polynomial $F(\theta) = \sum_n c_n e^{in\theta}$ is the polynomial slide with $x = e^{i\theta}$: multiplying two such responses **convolves their coefficient lists** c_n . That is the case we live in, because our response is a product.

Convolution	Multiplication
[1,2,3,4,5]	[1,2,3,4,5]
[10,9,8,7,6]	[6,7,8,9,10]
[6,19,40,70,110,114,106,85,50]	

The same two lists, convolved on the left and multiplied entry by entry on the right. A transform carries one picture into the other.

Our case: many nuclei, and the responses multiply

The electron is prepared in $|+\rangle$, a dynamical decoupling sequence runs, and the electron is read out at azimuth φ . The whole experiment is one number,

$$P(\varphi) = \frac{1}{2} + \frac{1}{2}|M| \cos(\varphi - \arg M), \quad M = \frac{1}{2} \text{Tr } V_0 V_1^\dagger,$$

where V_0 and V_1 are the propagators the **nuclear** spin experiences with the electron in $|0\rangle$ and in $|1\rangle$, exactly the M of chapter 1. If the two agree, $M = 1$ and no contrast is lost; a dip means the nucleus can tell the branches apart.

For several nuclei that do not interact with each other the trace factorizes,

$$M = \prod_j M_j, \quad M_j = \frac{1}{2} \text{Tr } V_0^{(j)} V_1^{(j)\dagger},$$

so each nucleus contributes its own response to the swept variable, and what we measure is their **pointwise product**. Contrast losses multiply, and for shallow dips they simply add, $1 - |M| \simeq \sum_j (1 - |M_j|)$, one comb of dips per nucleus as in chapter 1 (Taminiau et al., 2012).

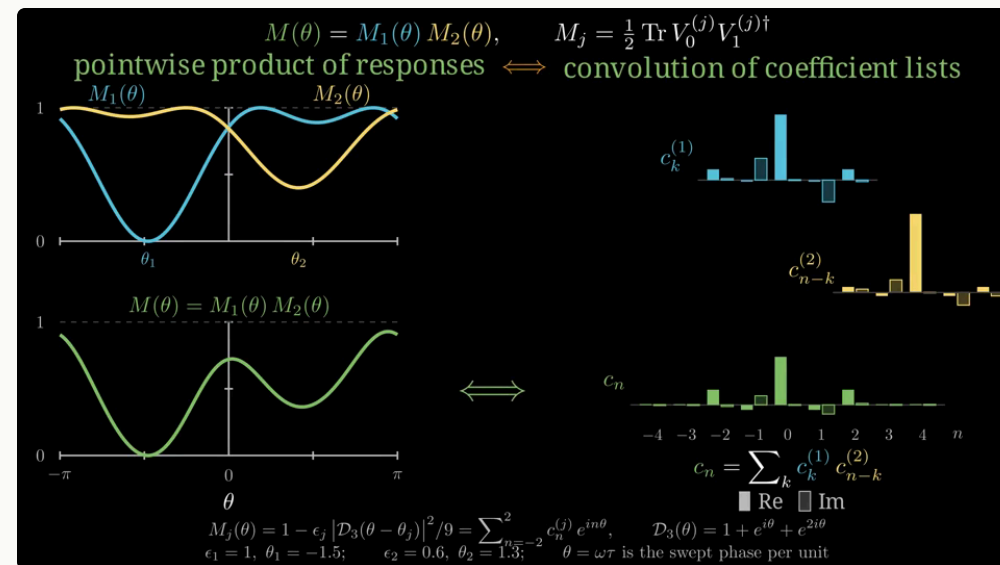
A product of responses is a convolution of coefficients

Our response is a product, $M = \prod_j M_j$. In Fourier space that product is a convolution: the coefficient lists of the single-nucleus responses are convolved, one more nucleus is one more convolution.

Each $M_j(\theta)$ is a function of the swept variable θ and, for a sequence of K equal units, a trigonometric polynomial $M_j(\theta) = \sum_n c_n^{(j)} e^{in\theta}$ of degree $K - 1$. Multiplying the responses is the polynomial slide with $x = e^{i\theta}$:

$$M_1 M_2 = \sum_n c_n e^{in\theta}, \quad c_n = \sum_k c_k^{(1)} c_{n-k}^{(2)} = (c^{(1)} * c^{(2)})_n.$$

The degrees add, the lists get longer, and the measured curve is the product of the dips. Chapter 3 turns this around: a QSP sequence multiplies degree-one factors on purpose, so it convolves its coefficient list one tap at a time.



Two nuclei give two dips $M_1(\theta)$ and $M_2(\theta)$; the measured response is their pointwise product, and its Fourier list is the convolution of the two coefficient lists (degree $2 + 2 = 4$). One more nucleus is one more convolution.

03

Quantum signal processing

The sequence is a polynomial, the polynomial is built by convolution, and the response is the transform of the taps

METHOD

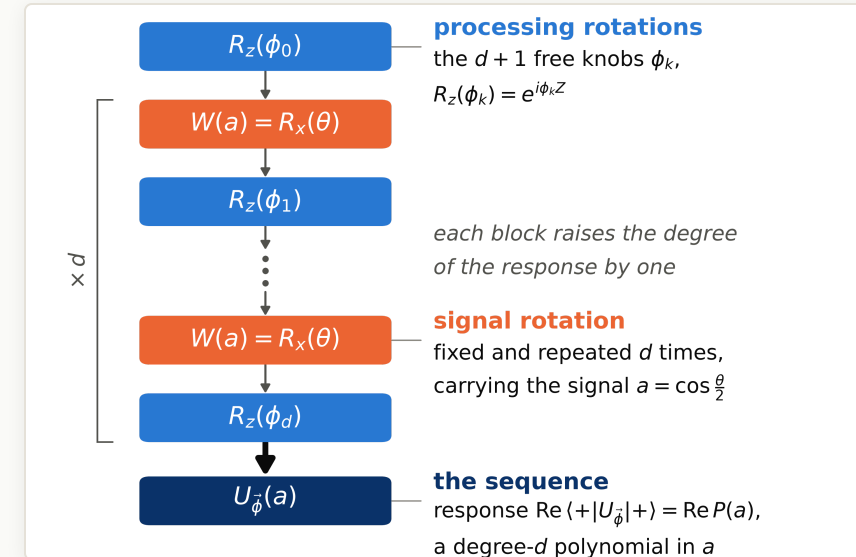
The sequence: the wait is the signal, the pulse is the knob

While the drive is off, the only surviving term is the detuning, so a wait of length τ is a precession about the z axis,

$$W(\delta) = R_z(\delta\tau) = \begin{pmatrix} z^{-1/2} & 0 \\ 0 & z^{1/2} \end{pmatrix}, \quad z = e^{i\delta\tau}.$$

It is the **signal**: the detuning δ is what the system hands us and W is the only place it enters. The microwave pulses G_k are the **processing**, chosen freely through their area β_k and carrier phase φ_k . With d equal waits and $d + 1$ pulses,

$$U(\delta) = G_d W(\delta) G_{d-1} \cdots W(\delta) G_0.$$



The textbook ladder of [1] puts the signal in an x -rotation $W_x(a)$ and the processing in z -rotations. It is the same sequence in a rotated basis, since $H W_x(a) H$ is our wait with $a = \cos(\delta\tau/2)$ and $H e^{i\phi_k Z} H = R_x(-2\phi_k)$ is our pulse.

Every factor is degree one, so the sequence is a polynomial

Split the half-powers off the wait, $W(\delta) = z^{-1/2} A(z)$ with $A(z) = \text{diag}(1, z)$, and collect the d of them into one overall phase:

$$U(\delta) = z^{-d/2} \tilde{U}(z), \quad \tilde{U}(z) = G_d A(z) G_{d-1} A(z) \cdots A(z) G_0.$$

Each factor $G_k A(z)$ is a 2×2 matrix whose entries are **degree-one polynomials in z** , so the product is a matrix polynomial of degree d , and d is simply the number of waits. Since every G_k and every W has unit determinant, $U(\delta) \in SU(2)$ and its first column is a unit vector:

$$U(\delta) = \begin{pmatrix} \hat{P}(\delta) & -\hat{Q}(\delta)^* \\ \hat{Q}(\delta) & \hat{P}(\delta)^* \end{pmatrix}, \quad |\hat{P}|^2 + |\hat{Q}|^2 = 1,$$

with $\hat{P} = z^{-d/2} P(z)$, $\hat{Q} = z^{-d/2} Q(z)$ and $\deg P, \deg Q \leq d$. We prepare $|0\rangle$ and read $|0\rangle$, so the experiment measures exactly $|\hat{P}(\delta)|^2$, and the transfer to $|1\rangle$ is $|\hat{Q}(\delta)|^2$. **Designing the sequence means designing one polynomial.**

Each added step convolves the coefficient list

The wait leaves the $|0\rangle$ component alone and advances the $|1\rangle$ component by one power of z , so

$A(z) = \Pi_0 + z \Pi_1$ with $\Pi_b = |b\rangle\langle b|$, and one step is a **two-tap kernel**

$$G_k A(z) = K_k^{(0)} + z K_k^{(1)}, \quad K_k^{(b)} = G_k \Pi_b.$$

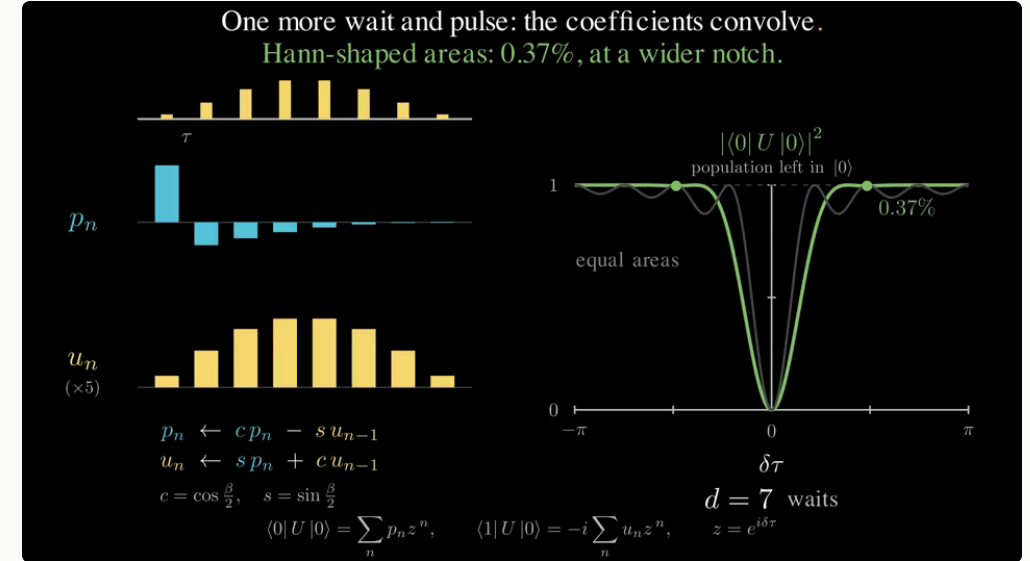
Write the state after k steps as a list of 2-vectors,

$\tilde{U}_k |0\rangle = \sum_n v_n^{(k)} z^n$. Multiplying by the new factor gives

$$v_n^{(k)} = K_k^{(0)} v_n^{(k-1)} + K_k^{(1)} v_{n-1}^{(k-1)} \equiv (K_k * v^{(k-1)})_n.$$

This is the identity of Section 2 with matrix-valued taps: a product of polynomials is a convolution of coefficients.

Starting from one tap $v_0^{(0)} = G_0 |0\rangle$, each step lengthens the list by exactly one, which is why **degree = number of waits**.



Left: the two coefficient lists of $\langle 0|U|0\rangle$ and $\langle 1|U|0\rangle$, each gaining one tap per wait and pulse. Right: the response $|\langle 0|U|0\rangle|^2$ over $\delta\tau$, which sharpens with the degree d and changes shape when the pulse areas are windowed.

The response is the Fourier transform of the taps

The signal variable lives on the unit circle, $z = e^{i\delta\tau}$, so the polynomial evaluated along the experiment's sweep axis is a Fourier series in the detuning,

$$\hat{P}(\delta) = \sum_{n=0}^d p_n e^{i(n-\frac{d}{2})\delta\tau}, \quad |\hat{P}(\delta)|^2 = \sum_{m=-d}^d r_m e^{im\delta\tau}, \quad r_m = \sum_n p_{n+m} p_n^*.$$

The measured curve is the transform of the **autocorrelation** of the tap list, which is the pointwise side of the one identity of Section 2: convolving the taps and multiplying the responses are the same statement read in two places. Two numbers follow immediately.

- The wait τ is the sampling interval, so the response repeats with period $2\pi/\tau$ in δ . Choosing τ is choosing where the lines of the spectrum land on the circle.
- The total time $T = d\tau$ is the record length, so the narrowest feature the response can have is $\Delta\delta \sim 2\pi/T$. **Resolution is bought with sequence length**, exactly as in Section 2.

RESULTS

Weak pulses: the taps are the pulse amplitudes

Expand each pulse to first order,

$$G_k \simeq I - \frac{i}{2} \beta_k (\cos \varphi_k X + \sin \varphi_k Y):$$

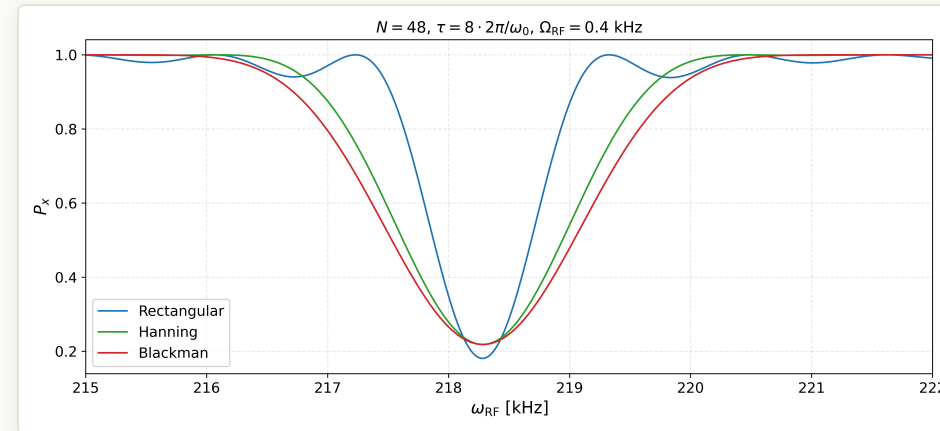
$$\hat{Q}(\delta) \simeq -\frac{i}{2} \sum_{k=0}^d \beta_k e^{i\varphi_k} e^{i(\frac{d}{2}-k)\delta\tau}.$$

The tap q_k is the complex pulse amplitude $\beta_k e^{i\varphi_k}$. Equal pulses give the all-ones boxcar of length $N = d + 1$, hence

$$|\hat{Q}(\delta)|^2 = \frac{(N\beta)^2}{4} |\mathcal{D}_N(\delta\tau)|^2, \quad \mathcal{D}_N(\theta) = \frac{\sin(N\theta/2)}{N \sin(\theta/2)},$$

the Dirichlet kernel that Section 2 met as the instrument function of a decoupling spectrum.

Shaping the β_k is apodization, so in this limit QSP is classical window design. At large pulse areas the linear map from amplitudes to taps is replaced by the exact matrix convolution we started from, and that is where QSP goes further.



The same trade in our own DDrF data: per-cell amplitudes $\Omega_k = \Omega f(k)$ flatten the detuned region at the cost of a wider resonance, at $N = 48$ cells.

(source: lab meeting 2026-04-28)

The QSP theorem: the reachable responses are known exactly

Theorem. With d waits and $d + 1$ pulses the sequence is always $U = \begin{pmatrix} \hat{P} & -\hat{Q}^* \\ \hat{Q} & \hat{P}^* \end{pmatrix}$ with $\deg P, \deg Q \leq d$ and $|\hat{P}|^2 + |\hat{Q}|^2 = 1$ on the circle, and conversely **every** such pair is realized by some choice of pulses [2, Thm. 3]. The textbook W_x form is the restricted version, with real coefficients and definite parity: $\deg P \leq d, \deg Q \leq d - 1$, parities d and $d - 1$, and $|P|^2 + (1 - a^2)|Q|^2 = 1$ [1, Thm. 1].

The forward direction is nothing but the convolution recursion: each step appends one tap and raises the degree by one, which is what flips the parity in the W_x form, while unitarity is inherited because every factor is unitary on the circle. The converse is the direction we use, because it says the design space is a **complete** list rather than a perturbative family: any bounded polynomial of degree d is a sequence we can actually build, so a polynomial that vanishes exactly at three chosen detunings is a gate that is exactly blind to three spectral lines.

What P is with no pulses, and what it can be with them

Switch every pulse off. The sequence is bare free precession, $U = W(\delta)^d$, whose tap list has a single nonzero entry, so its response is one pure harmonic,

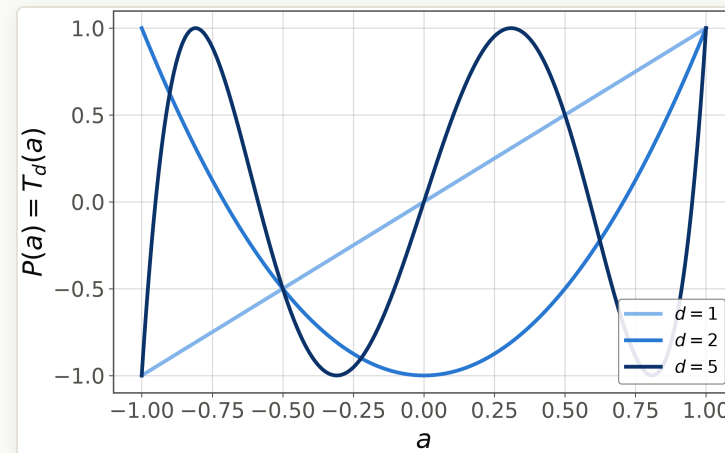
$$\langle 0|U|0\rangle = e^{-id\delta\tau/2}, \quad \text{Re } \langle 0|U|0\rangle = \cos(dx) = T_d(a),$$

with $a = \cos x$ and $x = \delta\tau/2$. In the textbook labelling this is the trivial-phase case of [1], $P = T_d(a)$ and

$Q = U_{d-1}(a)$, Chebyshev of the first and second kind, and the unitarity condition is just

$$T_d^2 + (1 - a^2)U_{d-1}^2 = \cos^2(dx) + \sin^2(dx) = 1.$$

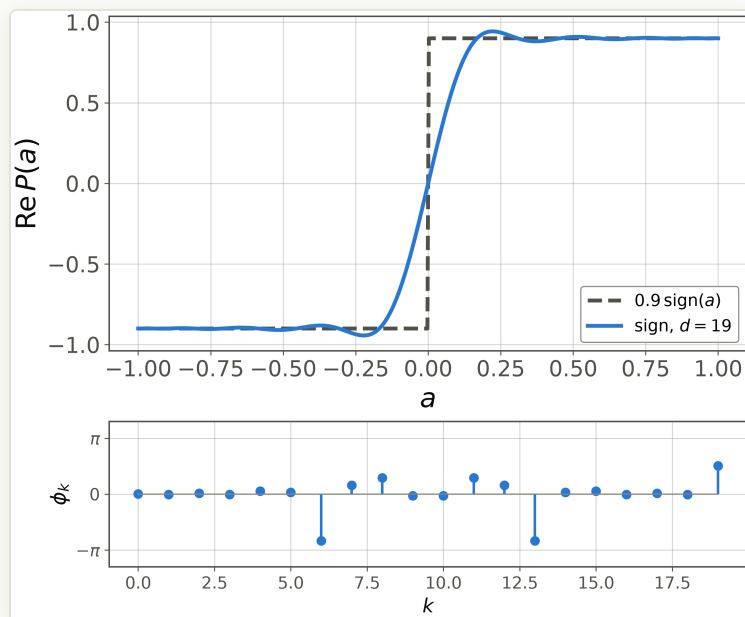
Chebyshev polynomials are the pure harmonics of the wait, and the pulses are what turns one harmonic into a designed sum of them.



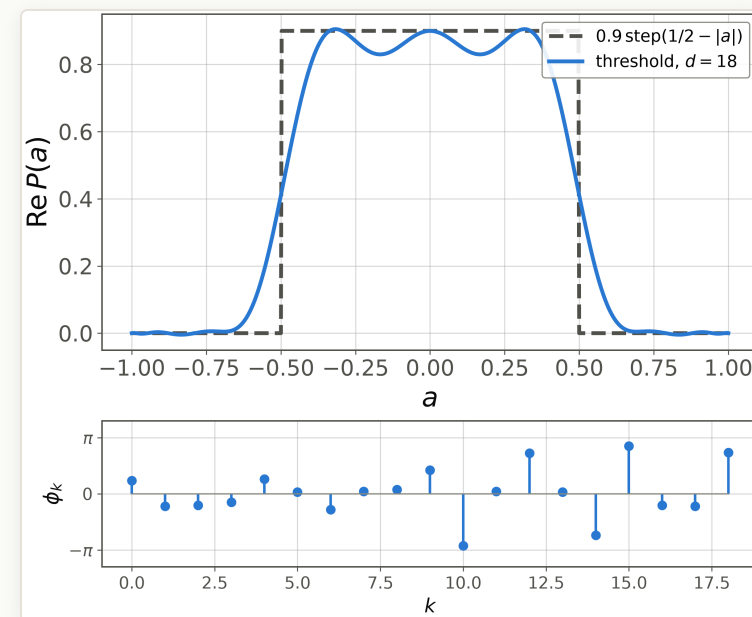
$P(a) = T_d(a)$ at trivial phases for $d = 1, 2, 5$, from our evaluator (agrees to 10^{-12}).

At $a = \pm 1$ the wait is trivial, so the sequence collapses to one fixed rotation and $|P(\pm 1)| = 1$ is forced: a target such as $\frac{1}{2} \text{sign}(a)$ is unreachable as P itself. Reading only the real part cures it, since $\text{Im } P$ can absorb the unit-modulus burden. **Re P may then be any real polynomial with parity $d \bmod 2$ and $|\text{Re } P| \leq 1$ [1].**

Choosing the polynomial is choosing the algorithm



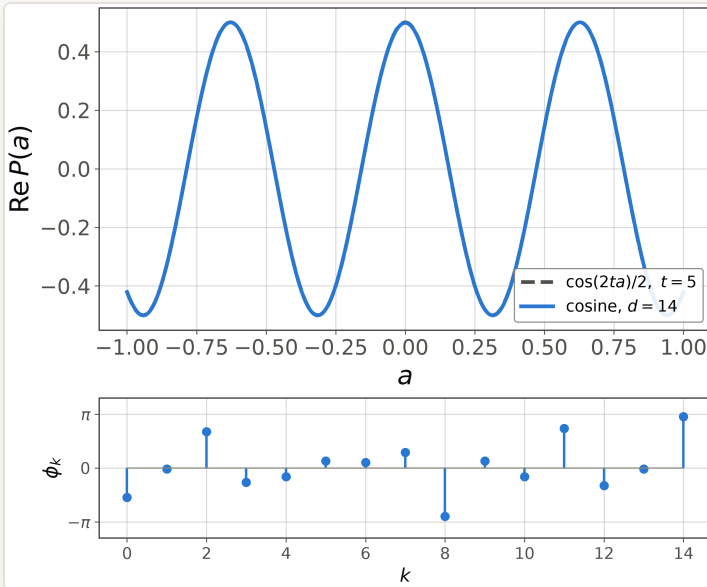
Sign function at $d = 19$, the decision primitive behind amplitude amplification and search.



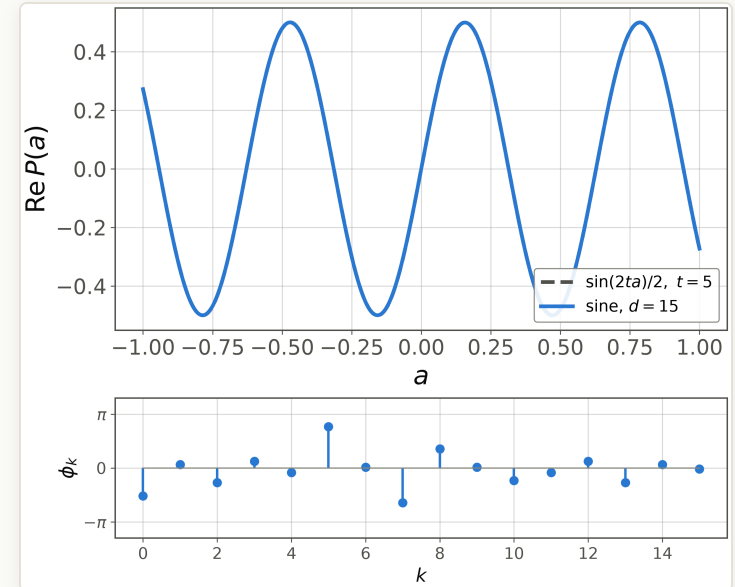
Threshold function at $d = 18$, a band-pass response used for eigenvalue thresholding and eigenstate filtering.

The lower panel of each figure is the phase list, which is the entire program. Appendix D of [1] gives lists for $\cos$, $\sin$, $1/a$ and $e^{-\beta a}$ as well.

Cosine and sine: one tap list is a simulation program



Cosine response at $t = 5$, $d = 14$, from the phase list of [1, App. D4].



Sine response at $t = 5$, $d = 15$, from the phase list of [1, App. D4].

Hamiltonian simulation uses the pair through $e^{-i\mathcal{H}t} = \cos(\mathcal{H}t) - i \sin(\mathcal{H}t)$, so these two angle lists are the entire program, and the degree grows linearly with the simulated time t .

STRATEGY

Finding the pulses is deconvolution

Given a target P with $|P| \leq 1$ on the circle, a partner Q with $|Q|^2 = 1 - |P|^2$ exists by the Fejér–Riesz factorization, and the pulses are then peeled off one at a time [2]. The two end taps of the matrix list are

$$C_d = G_d \Pi_1 G_{d-1} \Pi_1 \cdots \Pi_1 G_0, \quad C_0 = G_d \Pi_0 G_{d-1} \Pi_0 \cdots \Pi_0 G_0,$$

each containing a projector, hence each of rank one, and unitarity on the circle forces $C_d C_0^\dagger = 0$. Their column directions are therefore the two orthogonal columns $G_d|1\rangle$ and $G_d|0\rangle$: **the last pulse is read directly off the two ends of the tap list**. Removing it, $A(z)^{-1} G_d^\dagger \tilde{U}$ has degree $d - 1$, and d repetitions give every pulse. Dividing out one two-tap factor per step is the exact inverse of the step-by-step convolution that built the sequence.

Two practical alternatives exist. Optimization over the phases is what `pyqsp` and QSPACK do and what our own fits of 2026-08-18 did. For a **gate** target the whole design is even convex: maximizing the smallest projection of P on the target phase subject to $|P| \leq 1$ is a linear program with a global optimum.

Summary: three sections, one sentence

A quantum experiment prepares a state, applies a unitary and measures; the unitary is what we design, the measurement returns a response, and that response is a convolution that QSP lets us design exactly.

- Section 1 fixed the vocabulary: a state, a unitary built from a drift term and a control term, and a projective readout whose statistics are the response.
- Section 2 showed that responses multiply and therefore convolve, from the product over nuclear spins to the Dirichlet instrument function of a decoupling spectrum.
- Section 3 made the convolution the design variable. With d taps the sequence is a degree- d polynomial in $z = e^{i\delta\tau}$ assembled by d two-tap convolutions, the measured curve is the Fourier transform of that tap list, weak pulses reproduce classical windowing, and the theorem says every bounded polynomial is reachable with pulses recovered by a peel-off recursion.
- The concrete payoff is the conditional gate on the V_B^- centre, where a seven-pulse sequence is exactly the identity on three hyperfine lines and $R_x(\pi)$ on the fourth [3].

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7. Convolution and DDrf side peaks, lab meeting 2026-09-03 ([3. wiki/labmeeting/260903/](#)): the product-to-convolution derivation and the dice, polynomial and Fourier clips (rendered from 3Blue1Brown's *But what is a convolution?* source, MIT).
8. J. Haah, *Product decomposition of periodic functions in quantum signal processing*. Quantum **3**, 190 (2019); QSPPACK (Y. Dong *et al.*, Phys. Rev. A **103**, 042419 (2021)); pyqsp ([github.com/ichuang/pyqsp](#)).
9. New clips ([bloch-plus](#) , [seq-rabi](#) , [seq-ramsey](#) , [seq-odmr](#) , [bell-mixed](#) , [dd-cpmg](#) , [qsp-coeff-conv](#)) and the stepping figure: [3. wiki/code/lm-260923-animations/](#) , manim community 0.21, every state and axis computed from the actual matrices.

Thank you

Questions & discussion

Donghun Jung

Paulee Group, Center for Quantum Technology, KIST